

Unit 13: Worked Examples

1. Find the difference between each expression.

a. $(3x + 3.6) - (8.2x - 5)$

$$\begin{array}{r} 3x + 3.6 \\ - 8.2x + 5 \\ \hline -5.2x + 8.6 \end{array}$$

b. $(x - \frac{3}{4}) - (3\frac{1}{8} - x)$

$$\begin{aligned} x - \frac{3}{4} - 3\frac{1}{8} + x \\ x - \frac{6}{8} - 3\frac{1}{8} + x &= \boxed{2x - 3\frac{7}{8}} \end{aligned}$$

2. Are the following expressions equivalent? Show your work.

$\frac{9}{4} + (2.5 - 6\frac{1}{2})$ and $(-\frac{1}{2} \cdot \frac{3}{4} + 4.5) - 5\frac{7}{8}$

$$\frac{9}{4} + (2\frac{1}{2} - 6\frac{1}{2})$$

$$\frac{9}{4} - 4$$

$$\frac{9}{4} - \frac{16}{4} = -\frac{7}{4}$$

$$(-\frac{3}{8} + 4\frac{1}{2}) - 5\frac{7}{8}$$

$$(-\frac{3}{8} + \frac{9}{2}) - 5\frac{7}{8}$$

$$(-\frac{3}{8} + \frac{36}{8}) - 5\frac{7}{8}$$

$$\frac{33}{8} - \frac{47}{8} = -\frac{14}{8} = -\frac{7}{4}$$

Both produce the same answer and are therefore equivalent

3. Determine if the expressions are equivalent by substituting the same values for the variables.

$(5y - 3) - \frac{3}{4}(8 + 2y)$ and $(\frac{3}{2}y - 9 + 2y)$

$y = 2$

$$(10 - 3) - \frac{3}{4}(8 + 4)$$

$$7 - \frac{3}{4}(12)$$

$$7 - 9 = -2$$

$$[\frac{3}{2}(2) - 9 + 2(2)]$$

$$(3 - 9 + 4)$$

$$-2$$

The expressions are equivalent!